Checkpoint 4: Network Path Analysis Report

Student Name: Adeena Riaz and Ali Asad Khan
Date: 24/10/25
Target Host: www.apple.com (23.217.112.210)
Test Duration: ~2.4 hours
Total Packets Sent: 42,744
Total Packets Received: 39,060

Question 1: Overall Delivery Rate

Delivery Rate: 91.38% (0.9138)
Packet Loss Rate: 8.62% (0.0862)

The network path showed a delivery rate of 91.38%, meaning approximately 8.62% of packets were lost during the 2.4-hour test period. Out of 42,744 echo requests sent, 39,060 received replies. This indicates moderate packet loss, which is higher than typical stable connections but acceptable for Internet paths with variable conditions.

Question 2: Longest Success Streak

Answer: 182 consecutive successful pings

The longest period without any packet loss was 182 consecutive successful pings (approximately 36 seconds at 5 pings/second). This indicates that the network can maintain stable connectivity for short to moderate periods.

Question 3: Longest Loss Burst

Answer: 138 consecutive packet losses

The longest burst of consecutive packet losses was 138 packets (approximately 27.6 seconds). This significant loss burst suggests that when network issues occur, they tend to persist for extended periods rather than being isolated random losses.

Question 4: Loss Correlation Analysis

P(success | previous success): 0.9251 (92.51%)
P(success | previous loss): 0.7945 (79.45%)
Overall delivery rate: 0.9138 (91.38%)

Analysis: The probability of a successful ping following another successful ping (92.51%) is significantly higher than the probability of success following a loss (79.45%). The difference of approximately 13% indicates that **packet losses are CORRELATED and occur in bursts** rather than being independent random events.

This bursty behavior suggests that losses are likely caused by temporary network congestion or routing issues that affect multiple consecutive packets, rather than random interference affecting individual packets.

Question 5: Minimum RTT

Answer: 94.20 ms

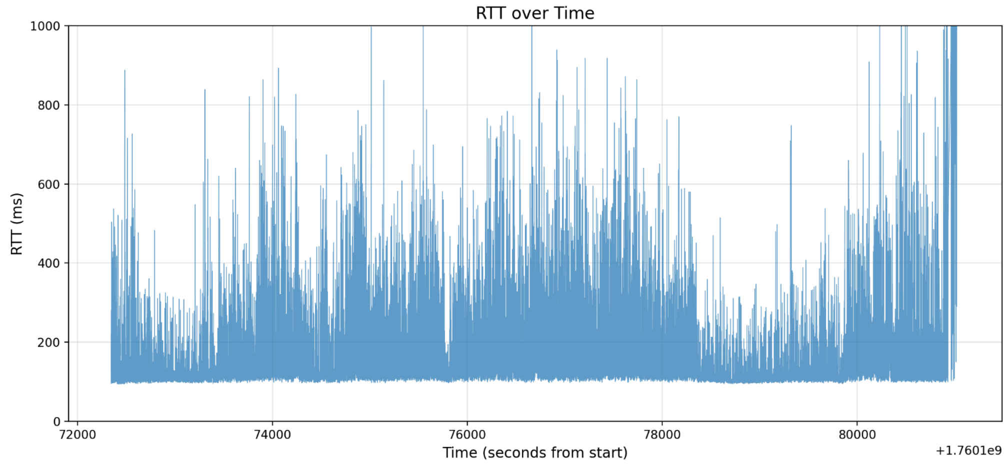
The minimum observed RTT of 94.20 ms represents the best-case latency for this network path under ideal conditions with no congestion. This is likely close to the true propagation delay plus minimal processing time.

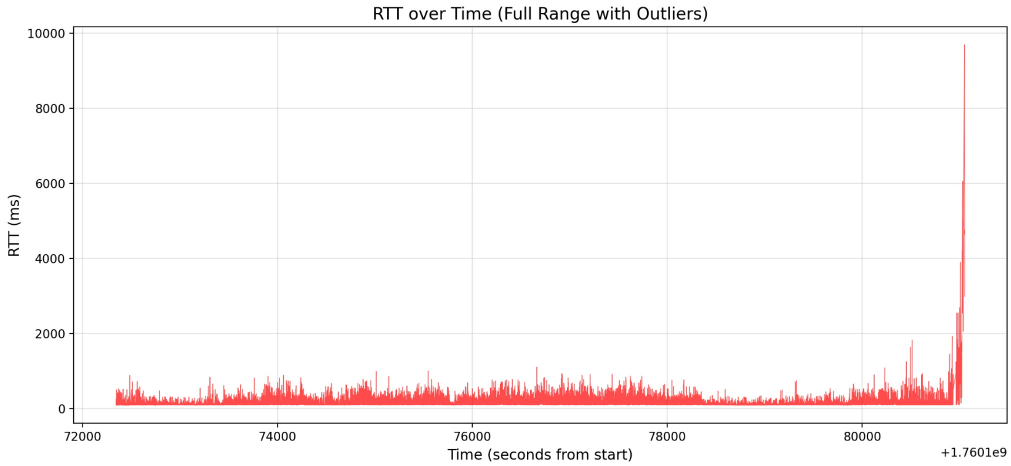
Question 6: Maximum RTT

Answer: 9,692 ms (9.69 seconds)

The maximum RTT observed was 9.69 seconds, which represents a severely delayed packet or near-timeout condition. This extreme value occurred during a period of significant network congestion or routing problems.

Question 7: RTT vs Time Graph



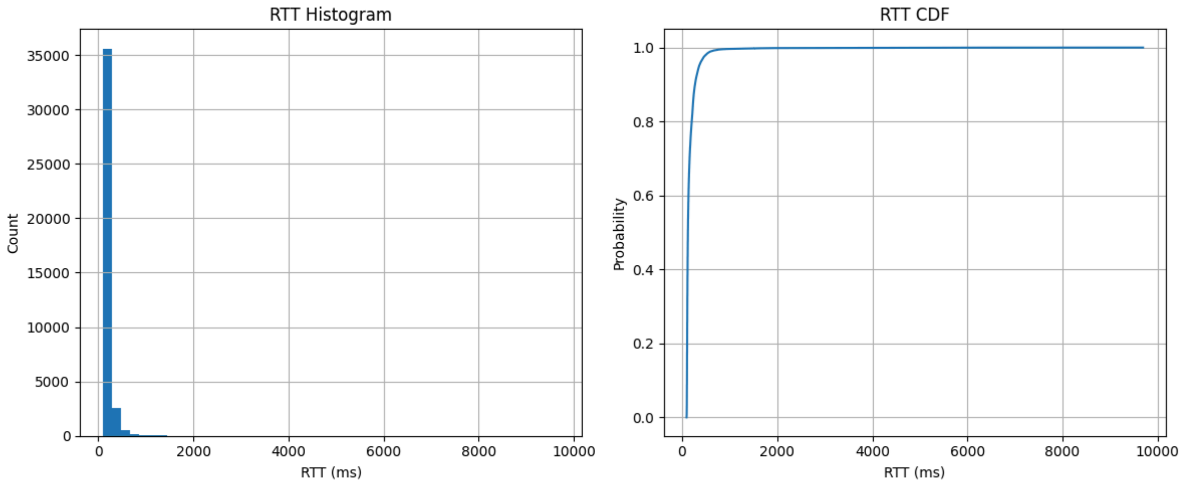


The RTT vs Time graph shows several interesting patterns:

- **Baseline RTT:** Most of the time, RTT fluctuates between 100-300 ms
- **Periodic spikes:** Regular spikes up to 600-1000 ms suggest periodic network congestion
- **Stable periods:** There are periods where RTT remains relatively stable around 100-150 ms
- **Congestion periods:** Mid-test and end of test show increased variability and higher RTTs
- **Time variation:** The network behavior changes over time, likely due to varying traffic loads

---

Question 8: RTT Distribution



Statistics:

- **Mean RTT:** 170.05 ms
- **Median RTT:** 123.00 ms
- **Range:** 94.20 - 9692.00 ms

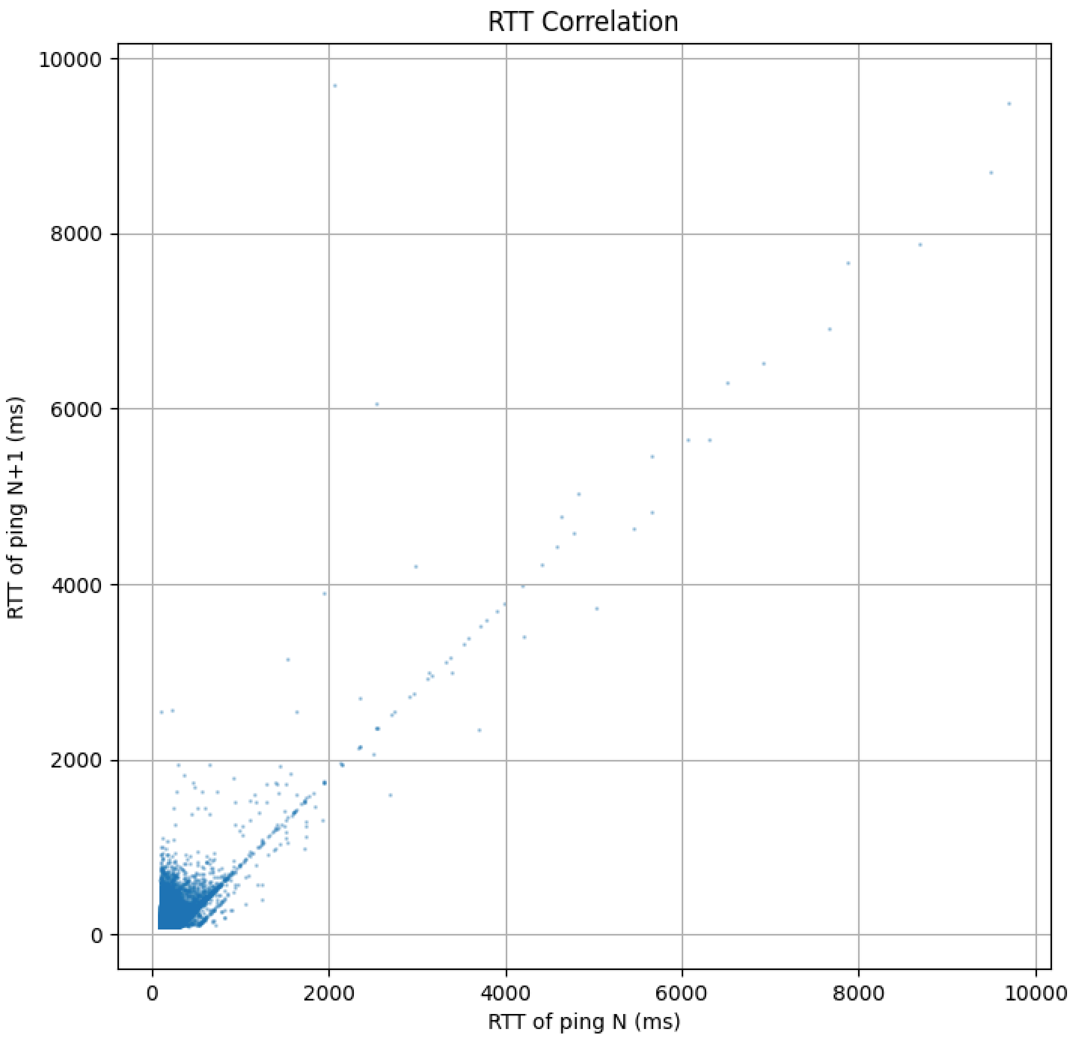
Distribution Shape: The histogram shows a **highly skewed right distribution** with:

- Large concentration of values around 100-200 ms (modal peak)
- Long right tail extending to several seconds
- The distribution is NOT normal - it's heavily skewed with outliers

The CDF shows a sharp initial rise, indicating that most packets (>80%) complete within 100-400 ms, but the long tail indicates occasional severe delays.

The mean (170 ms) being higher than the median (123 ms) confirms the right skew - a few very large RTT values pull the mean up.

Question 9: RTT Correlation



Correlation Coefficient: 0.8361

Analysis: The correlation coefficient of 0.8361 indicates **strong positive correlation** between consecutive RTT measurements. This means:

- If ping N has high RTT, ping N+1 is likely to also have high RTT
- If ping N has low RTT, ping N+1 is likely to also have low RTT
- Network conditions change slowly relative to the 0.2-second ping interval

The scatter plot shows:

- Dense clustering along the diagonal in the lower-left corner (100-400 ms range)
- Points near the diagonal indicate strong correlation
- Some outliers represent transition periods between good and congested states
- The strong correlation suggests network congestion/conditions persist for multiple consecutive packets

---

Question 10: Conclusions

Network Behavior Analysis

Expected Behavior: I expected the network path to show relatively stable RTT with occasional spikes due to Internet congestion, and some packet loss due to the long distance to Apple's servers.

Actual Observations:

1. **Packet Loss Pattern:** The 8.62% loss rate was higher than expected for a major service like Apple. The strongly correlated losses (Question 4) indicate network issues occur in bursts rather than randomly, suggesting congestion or routing instabilities.

2. **RTT Stability:** The network showed moderate variability with a baseline around 100-150 ms and frequent spikes to 300-1000 ms. The strong correlation (0.8361) between consecutive RTTs confirms that network conditions change slowly.

3. **Periodic Congestion:** The RTT vs Time graph reveals clear periods of increased congestion, particularly in the middle and end of the test. This could be due to:

   - Time-of-day traffic patterns
   - Network routing changes
   - Intermediate router congestion

Surprises

Most Surprising Findings:

1. **High Loss Burst Duration:** The longest loss burst of 138 consecutive packets (27.6 seconds) was surprisingly long. This suggests a significant routing issue or path failure during that period.

2. **Extreme RTT Outliers:** The maximum RTT of 9.69 seconds indicates packets that nearly timed out, which is unusual for a major CDN like Akamai (Apple's content delivery network).

3. **Variability Over Time:** The clear variation in network behavior across different time periods was more pronounced than expected, with distinct "good" and "bad" periods visible in the time series.

Network Path Characteristics

Based on the data, this network path exhibits:

- **Moderate reliability** (91.38% delivery)
- **Bursty loss pattern** (correlated failures)
- **Variable latency** (high standard deviation)
- **Strong temporal correlation** (conditions persist across packets)

The path appears to be subject to periodic congestion, possibly at intermediate routers between Pakistan and Apple's CDN servers. The bursty nature of losses and strong RTT correlation suggest that network issues, when they occur, affect multiple consecutive packets rather than individual packets randomly.

Test Configuration

Command Used:

ping -D -n -i 0.2 www.apple.com | tee data.txt

Test Parameters:

- Ping interval: 0.2 seconds (5 pings/second)
- Duration: ~2.4 hours (8,549 seconds)
- Total packets: 42,744
- Received: 39,060
- Lost: 3,684

---

adeenaKhan@ubuntu24: ~/Desktop/minnow-Stanford

```
adeenaKhan@ubuntu24:~/Desktop$ cd minnow-Stanford
adeenaKhan@ubuntu24:~/Desktop/minnow-Stanford$ mtr --report www.apple.com
Start: 2025-10-11T13:43:40+0500
HOST: ubuntu24                    Loss%   Snt   Last   Avg  Best  Wrst StDev
  1.|-- a23-217-112-210.deploy.st 30.0%    10  102.2 272.7  98.4 1077. 359.4
adeenaKhan@ubuntu24:~/Desktop/minnow-Stanford$
```